

Omer Yuval

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Computational Neuroscientist & Software Engineer (PhD) specializing in reinforcement learning, physical simulations, and full-stack development. I architect and lead end-to-end computational pipelines—from computer vision and automated object tracking to dynamic modeling and RL-driven simulations. With a proven track record of translating complex mechanical intelligence concepts into robust software and insight, I successfully manage cross-disciplinary projects from conceptualization to peer-reviewed delivery. Experienced in guiding individual researchers and designing physical experiments to bridge real-world data with simulations, I leverage a technical toolkit spanning Python, MATLAB, and JavaScript (React, Node.js) to optimize decision-making and build scalable solutions.

Education and employment

Postdoctoral researcher (April 2023 - current)

- Faculty of Life Sciences, Tel Aviv University, Israel.
- Subject: Mechanical modelling of insect control of locomotion using model-free reinforcement learning.

Research assistant in Intelligent Transport Systems (July 2022 - November 2022)

- Institute for Transport Studies, University of Leeds, UK.
- Subject: Multi-objective optimisation of vehicle speed profile for reducing emission and fuel consumption.
- Part-time.

Teaching online programming lessons for kids (April 2022 - April 2023)

- Cypher Coders, UK.
- Subject: Using javascript to develop 2D and 3D games.
- Part-time.

PhD in computational neuroscience (2017- January 2022)

- School of Computing, Faculty of Engineering, University of Leeds, UK.
- Subject: The neuromechanical mechanisms underlying the locomotion of the microswimmer *C. elegans* in 3D environments.
- University funded.

Master's degree in computational neuroscience (2014 - 2016)

- Faculty of biology, Technion, Haifa, Israel.
- Subject: Segmentation and morphological analysis of a highly-branched neuron in *C. elegans*, used to study the interconnection between neuronal structure and function.

- University funded.

Bachelor's degree in biology (2010 - 2014)

- Faculty of biology, Technion, Haifa, Israel.

Experience

- **Programming languages:** Python, Javascript, MATLAB, HTML, PHP and SQL.
- **Full stack development:** React, CSS, Node, Express, MongoDB, Docker, Cache (Redis).
- **Projects & skillset:** Mechanical modelling, Reinforcement learning, Navigation, 3D Computer vision, Image segmentation, 3D Object tracking, multi-objective optimisation, Mathematical modelling of neuronal dynamics, Motor control, Multi-camera calibration, Parallel computing, HPC (CPU/GPU), Linux, Windows, Graphical user-interface.
- **Teaching assistant:** Machine learning (Python), Procedural programming (C), Object-oriented programming (Java), Bioinformatics practicals (python and statistics), Experimental skills in neuroscience (Java and fiji/imageJ), Intermediate Skills for Professional and Academic Development (C++).
- **Laboratory work:** 3D imaging and calibration, Confocal microscopy, Locomotion and navigation assays, Optogenetics, Calcium-imaging.

Publications

- **Yuval, O.**, Ozeri, E., Amir, A., & Ayali, A. (2026). Biologically Constrained Insect Models Enable Realistic Simulations and Improve Biomechanical Predictions of Locomotion. *Soft Robotics*, 21695172261435551. <https://doi.org/10.1177/21695172261435551>.
- **Yuval, O.**, Amir, A., & Ayali, A. (2026). Multiple locomotion gaits in the mole cricket. *Journal of Bionic Engineering*, 23(2), 863-879. <https://doi.org/10.1007/s42235-025-00838-x>.
- **Yuval O.**, Amir A., Ozeri E., Liti L., Ayali A. (2025). Simulating locomotion under anatomical and mechanical constraints. Accepted for presentation at AMAM/LokoAssist, TU Darmstadt, Germany (to appear).
- Amir A., **Yuval O.**, Ayali A. Six-legged bound: a newly described insect gait (2025). *Royal Society Open Science*, 12(6), 250143. <https://doi.org/10.1098/rsos.250143>.
- Iosilevskii, Y., **Yuval, O.**, Shemesh, T., & Podbilewicz, B. (2024). Protocol for neuron tracing and analysis of dendritic structures from noisy microscopy images using Neuronalyzer. *STAR protocols*, 5(2), 103063. <https://doi.org/10.1016/j.xpro.2024.103063>.
- Ilett, T. P., **Yuval, O.**, Ranner, T., Cohen, N., & Hogg, D. C. (2023). 3D shape reconstruction of semi-transparent worms. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (pp. 12565-12575).
- **Yuval, O.**, Lin, Z., & Chen, H. (2023). Multiobjective speed profile optimisation considering fuel and NOx. In *Proceedings of 15th ITS European Congress*. Leeds.

- **Yuval, O.** (2022). The neuromechanical control of *Caenorhabditis elegans* head motor behaviour in 3D environments (Doctoral dissertation, University of Leeds).
- **Yuval, O.**, Iosilevskii, Y., Meledin, A., Podbilewicz, B., & Shemesh, T. (2021). Neuron tracing and quantitative analyses of dendritic architecture reveal symmetrical three-way-junctions and phenotypes of *git-1* in *C. elegans*. *PLoS Computational Biology*, 17(7), e1009185.
<https://doi.org/10.1371/journal.pcbi.1009185>.
- Deng, L., Denham, J. E., Arya, C., **Yuval, O.**, Cohen, N., & Haspel, G. (2021). Inhibition underlies fast undulatory locomotion in *Caenorhabditis elegans*. *Eneuro*, 8(2).
<https://doi.org/10.1523/ENEURO.0241-20.2020>.
- Salfelder, F., **Yuval, O.**, Illett, T. P., Hogg, D. C., Ranner, T., & Cohen, N. (2020). Markerless 3D spatio-temporal reconstruction of microscopic swimmers from video. *visual observation and analysis of vertebrate and insect behavior*.

Under review and in preparation

- **Yuval O.**, Amir A., Fuxman K., Ayali A. Dynamic control principles for adaptive gait transitions. Under review.
- **Yuval O.**, Amir A., Ben Zion, M. Y., Ayali A. Vibration of a tilted plane reveals passive mechanical tendency of insects for uphill body alignment and climbing. Manuscript in preparation.
- **Yuval O.**, Illett T., Salfelder F., Holbrook R., Runner T., Cohen N. The neuromechanical control of *C. elegans* head motor behaviour in 3D environments. Manuscript in preparation.
- **Yuval O.**, Cohen N. Bistable head motor neurons underlie spontaneous gait selection during chiral forward locomotion. Manuscript in preparation.

Reviews

- Peer reviewer for **Entomological Science**, 2024.
- Peer Reviewer for **The IEEE International Conference on Robotics and Biomimetics (ROBIO)**, August 2024.

Other projects

- **Semantic representation and matching of LaTeX expressions.**
Demo: <https://omer1yuval1.github.io/LaTeXs/>.

Conferences and demonstrations

- Poster presentation at the **AMAM/LokoAssist** at TU Darmstadt, Germany (2025).
- Talk at the **27th International Congress of Entomology** at Kyoto, Japan (ICE2024, 2024).
- Poster presentation at the **European Worm Meeting** (2020 and 2022).
- Poster presentation at the **UK Worm Meeting** at Leeds University (2018) and Imperial College London (2019).
- Poster presentation at the **International Worm Meeting** at UCLA (2015 and 2019).

- Poster presentation at the **UK Computational Worm Meeting** at The Royal Society, London, UK (2018).
- Conference presentation at the **annual Biology faculty retreat** in Israel (2014).
- Laboratory demonstrations for master's students in biology (2018-2019).
- Organisation of a stall for an **outreach event** at Leeds city museum (2019).
- Participation in a **fire-fighting robot contest** in Israel (2006 and 2000).